



**UNIVERSITI TEKNOLOGI MARA
FINAL TEST**

COURSE : MOBILE APPLICATION DEVELOPMENT
COURSE CODE : CSC660
SEMESTER : MARCH – AUGUST 2026
TIME : 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts:
PART A (15 Questions)
PART B (4 Questions)
2. Answer ALL questions in English. Start each answer on a new page.
3. Upload the file at the specified platform.
4. You are strictly prohibited to discuss/share/disseminate the questions and answers amongst your classmates/coursemates. If found guilty or committed to one of the actions, you will be penalised.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (15 MARKS)

Determine whether the following statements are **TRUE** or **FALSE**.

1. According to mobile UX design principles, approximately 94% of mobile phone usage occurs in portrait orientation, so mobile apps should always be designed portrait-first.
2. In the Thumb Zone concept, the top corners of a mobile screen are considered the "Green Zone" where primary action buttons should be placed.
3. The minimum recommended touch target size in iOS design guidelines is 44×44 points, because a finger is less precise than a mouse cursor.
4. The Gestalt Law of Closure explains why users can recognise an incomplete circle or logo as a complete shape, as the brain automatically fills in the missing parts.
5. According to Nielsen's usability heuristics, an app should keep users informed about what is happening through appropriate feedback within a reasonable time.
6. In a Flutter project, the pubspec.yaml file is used to declare app dependencies and assets, and the flutter pub get command must be run after editing it.
7. Hot reload (pressing 'r') in Flutter can detect and apply changes made to the pubspec.yaml file without restarting the app.
8. In Dart, a variable declared with the const keyword can be re-assigned a new value at runtime.
9. The main() function is the entry point of every Flutter application, and runApp() is called inside it to launch the root widget.
10. In Flutter, enabling Material Design 3 requires setting useMaterial3: true inside the ThemeData of the MaterialApp.
11. If a state variable is changed outside of setState(), the variable's value will change but the user interface will not refresh to show the new value.
12. A StatefulWidget in Flutter requires two classes: the widget class itself and a separate State class created through the createState() method.
13. The correct way to display a SnackBar in modern Flutter is by calling Scaffold.of(context).showSnackBar().
14. An AlertDialog blocks user interaction with the rest of the screen until the user responds, making it suitable for critical confirmations such as deleting data.
15. In an AlertDialog, Navigator.pop(context) is used to close the dialog and return to the previous screen.

PART B (55 MARKS)

Answer ALL questions.

QUESTION 1 [10 MARKS]

A design team is building a mobile e-wallet app called **"DuitPay"** for Malaysian users. The main screen must display the wallet balance, a "Pay" button, a "Top Up" button, and a transaction history list. The team wants the app to be easy to use with one hand and intuitive for first-time users.

- (a) Describe [C2] the Thumb Zone concept in mobile UX design. State where the "Pay" button should be placed on the DuitPay main screen based on this concept, and give ONE reason.

(3 marks)

- (b) Explain [C2] the following TWO Gestalt principles: (i) Law of Proximity and (ii) Law of Similarity. For EACH principle, give ONE example of how it can be applied in the DuitPay transaction history list.

(4 marks)

- (c) List [C1] THREE principles of effective mobile screen design that the DuitPay team should follow. For EACH principle, state ONE reason why it matters for an e-wallet app.

(3 marks)

QUESTION 2 [15 MARKS]

A junior developer has just installed Flutter and is building his first app for a campus event called **"MyConvo"**. The app displays the convocation schedule and a countdown. He faces several issues during development.

- (a) The developer runs the command flutter doctor and sees a [X] symbol next to "Android toolchain". Explain [C2] what flutter doctor does, and state TWO possible causes of this error.

(4 marks)

- (b) The developer writes the following Dart code:

```
void main() {  
  String uni = 'UiTM';  
  final String campus = 'Kuala Terengganu';  
  const int year = 2026;  
  var event = 'Convocation';  
  
  campus = 'Dungun';      // LINE X  
  event = 'Graduation';  // LINE Y  
  
  print('$event $year at $uni $campus');  
}
```

Identify [C1] which line (LINE X or LINE Y) will cause a compile error. Explain [C2] why that line causes the error and why the other line is valid.

(5 marks)

- (c) The developer adds an image logo.png into the assets/images/ folder, but the app shows a "Unable to load asset" error when running. State [C1] TWO steps he must take to fix this error.

(3 marks)

- (d) Differentiate [C2] between hot reload (r) and hot restart (R) in Flutter. State ONE situation where hot reload alone is NOT sufficient.

(3 marks)

QUESTION 3 [15 MARKS]

A student is building a Flutter app called **"WaterTracker"** that helps users record how many glasses of water they drink daily. The main screen displays the current count in large text, a button to add one glass, and a button to reset the count to zero.

- (a) Determine [C2] whether the WaterTracker main screen should be a StatefulWidget or a StatelessWidget. Justify your answer by explaining how the screen content is expected to behave.

(3 marks)

- (b) The student writes the following code for the "Add Glass" button, but when the button is tapped, the number on screen does not change even though the variable increases:

```
int glasses = 0;

ElevatedButton(
  onPressed: () {
    glasses++;
  },
  child: Text('Add Glass'),
)
```

Identify [C1] the cause of this problem and write [C3] the corrected code for the onPressed callback so that the UI updates correctly.

(5 marks)

- (c) Write [C3] the complete Dart code for the onPressed event handler of the "Reset" button. When tapped, the code must: (i) reset the glasses variable to 0 using setState(), and (ii) display a SnackBar with the message "Counter reset!" that automatically disappears after 2 seconds.

(7 marks)

QUESTION 4 [15 MARKS]

A Flutter app called **"NotaKu"** allows students to save and delete lecture notes. The developer must add a confirmation feature: when the user taps the "Delete Note" button, the app must ask for confirmation before deleting.

- (a) Compare [C4] `SnackBar` and `AlertDialog` as user feedback widgets in Flutter based on: (i) position on screen, (ii) whether it blocks user interaction, and (iii) ONE example of when each should be used in the `NotaKu` app.

(6 marks)

- (b) Write [C3] the complete Dart code for the `onPressed` event handler of the "Delete Note" button. The code must display an `AlertDialog` with the following requirements:

- The dialog title is "Confirm Delete".
- The dialog content text is "Delete this note permanently?".
- A "CANCEL" `TextButton` that closes the dialog without any action.
- A "DELETE" `ElevatedButton` with a red background that closes the dialog.

(9 marks)

END OF QUESTION PAPER